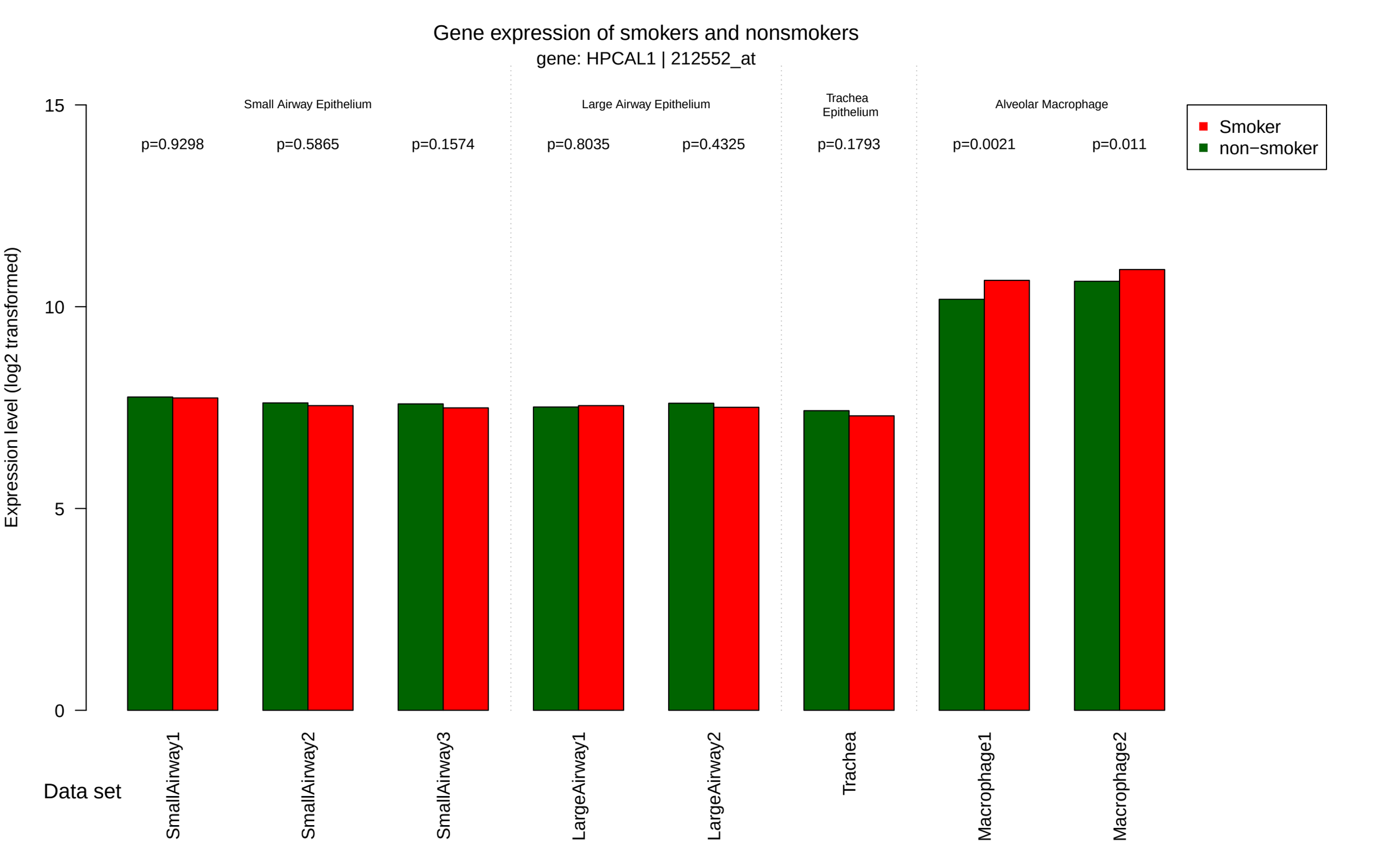
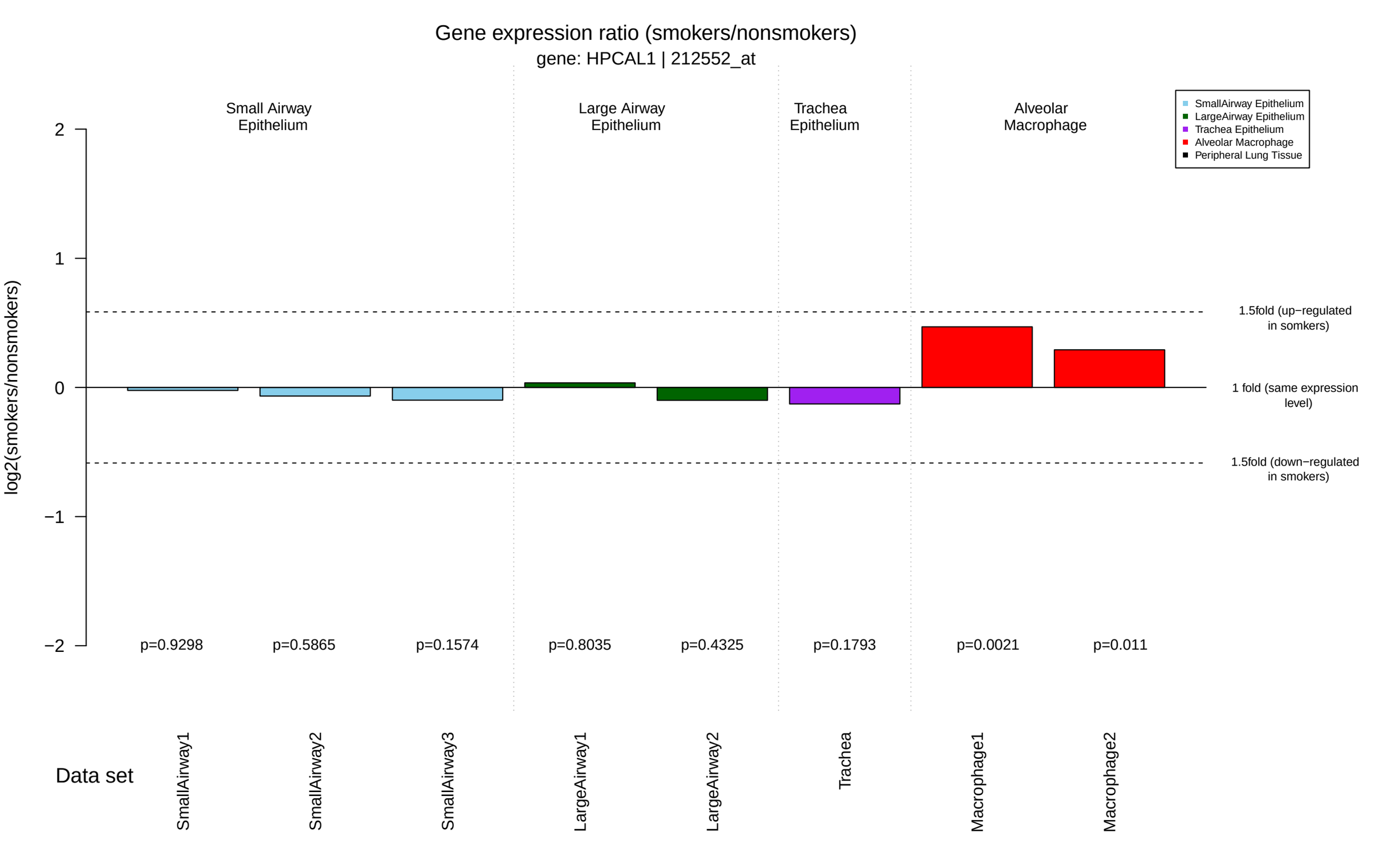
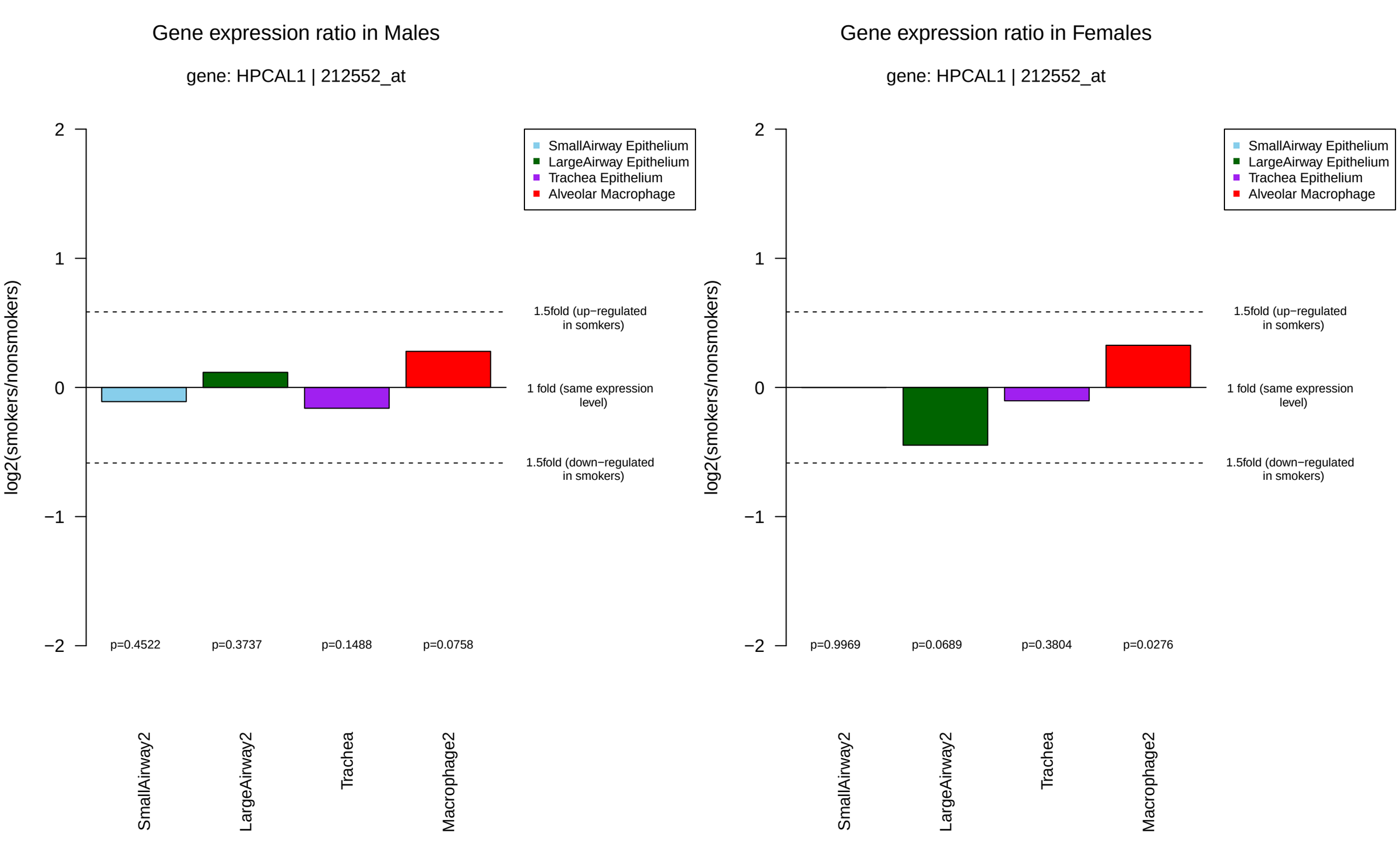
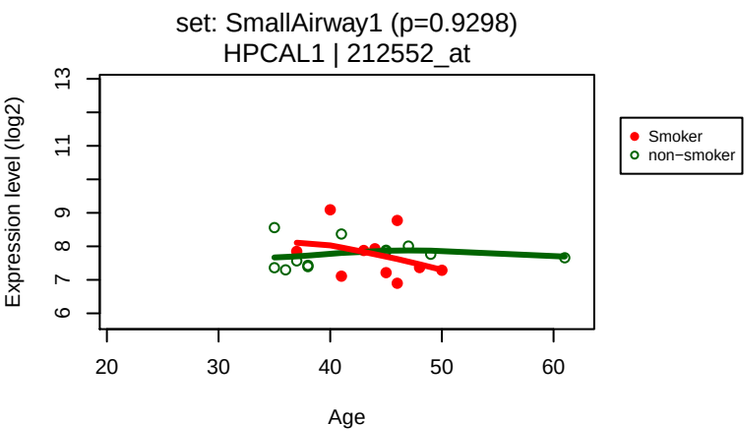
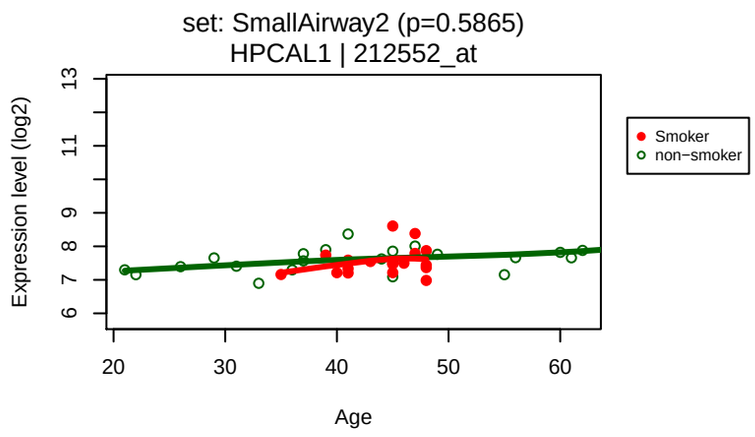
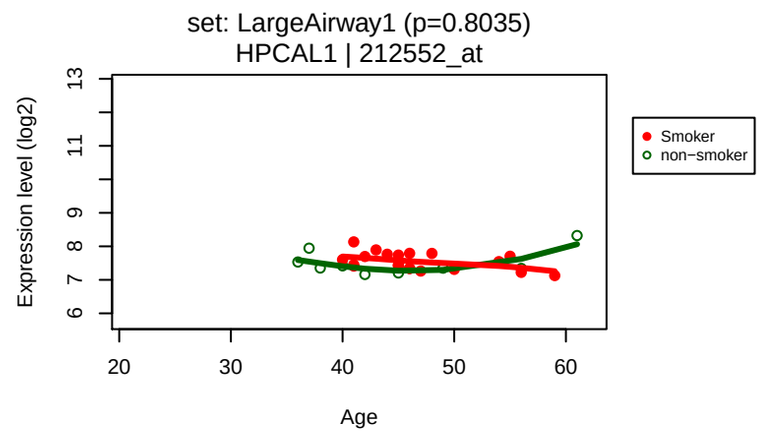

Gene expression of smokers and nonsmokers
gene: HPCAL1 | 212552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9298
p=0.5865
p=0.1574
p=0.8035
p=0.4325
p=0.1793
p=0.0021
p=0.011
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

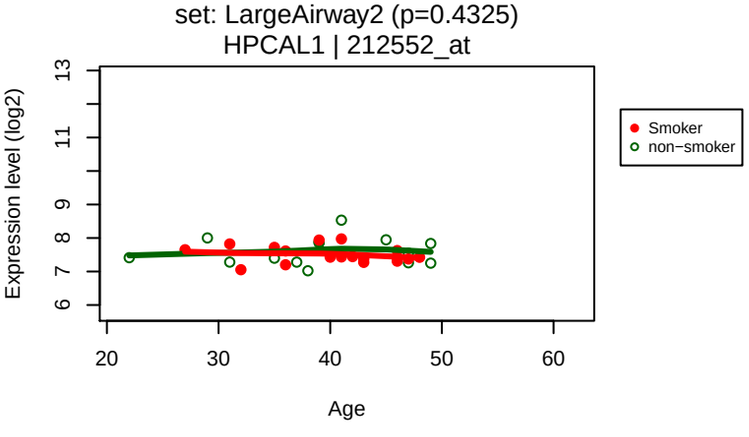

Gene expression ratio (smokers/nonsmokers)
gene: HPCAL1 | 212552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9298
p=0.5865
p=0.1574
p=0.8035
p=0.4325
p=0.1793
p=0.0021
p=0.011
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

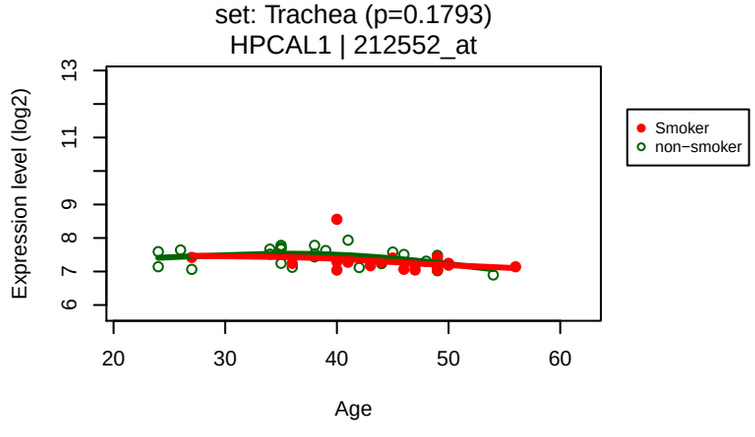

Gene expression ratio in Males
gene: HPCAL1 | 212552_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4522
p=0.3737
p=0.1488
p=0.0758
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPCAL1 | 212552_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9969
p=0.0689
p=0.3804
p=0.0276
SmallAirway2
LargeAirway2
Trachea
Macrophage2

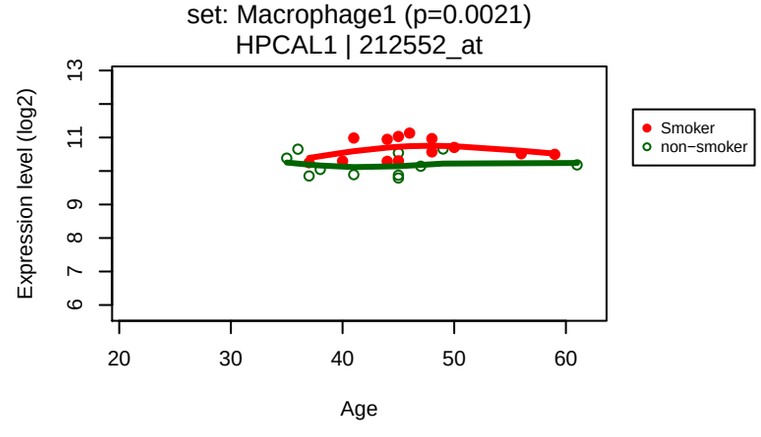

set: SmallAirway1 (p=0.9298)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker

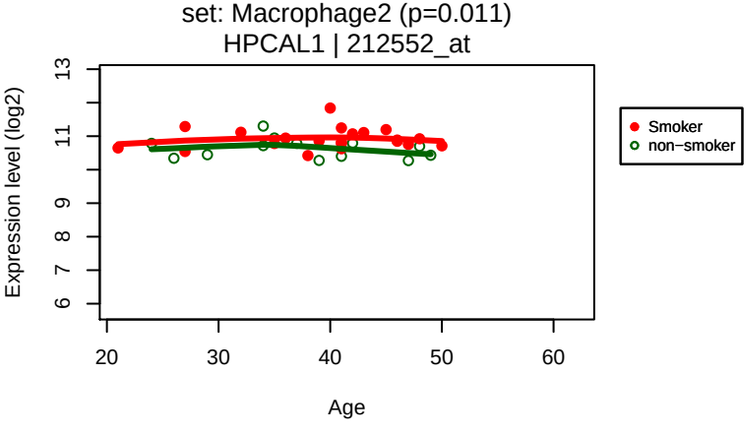

set: SmallAirway2 (p=0.5865)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker

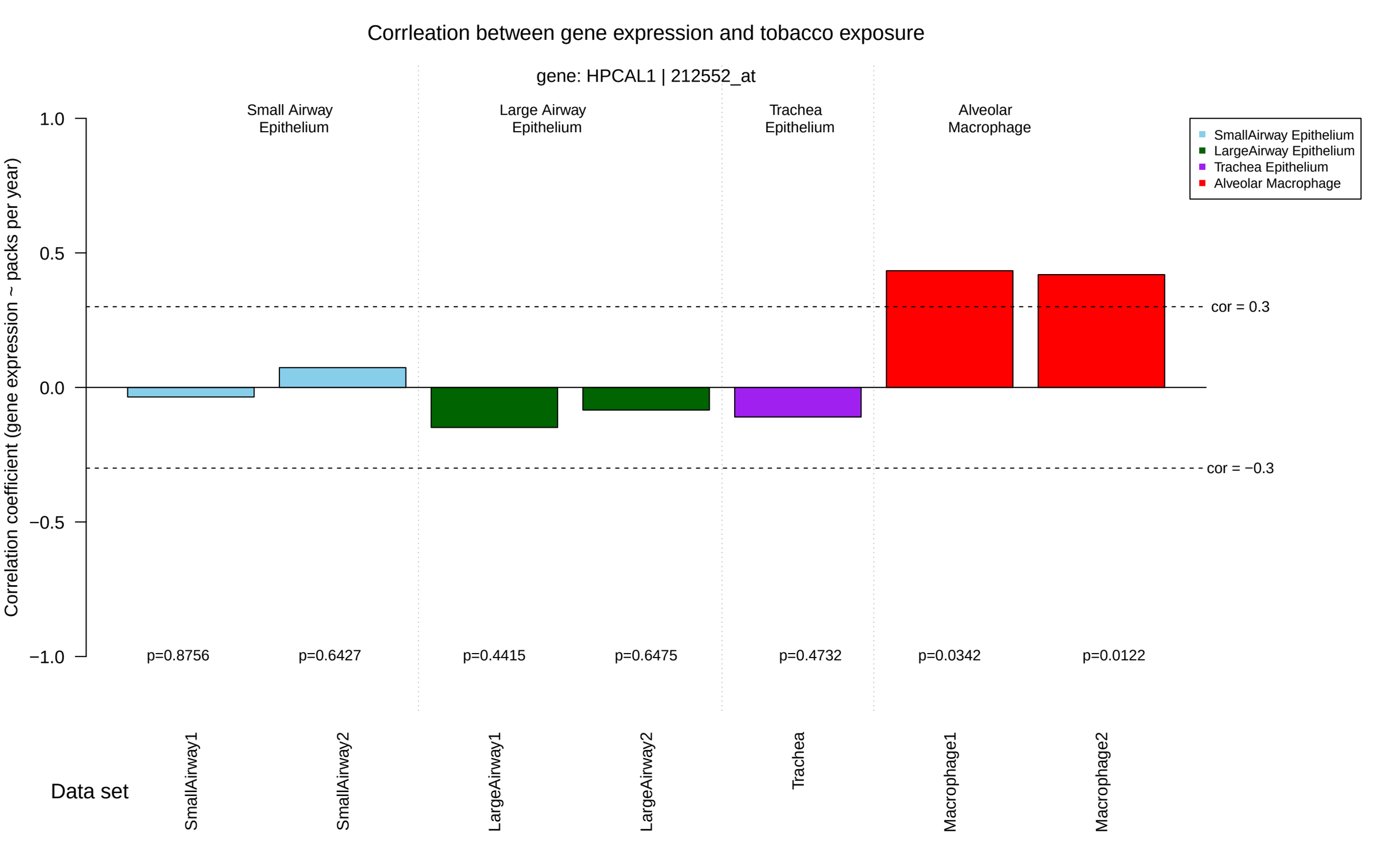

set: LargeAirway1 (p=0.8035)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4325)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1793)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0021)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.011)
HPCAL1 | 212552_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPCAL1 | 212552_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8756
p=0.6427
p=0.4415
p=0.6475
p=0.4732
p=0.0342
p=0.0122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2